

Supplementary Material

Fluoride Varnish in a Changing Preventive Environment: A Reanalysis and Temporal Meta-Regression of Historical and Contemporary Trials

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This Supplement provides the methodological details, data reconstruction, robustness checks, sensitivity analyses, diagnostics, and computational record supporting the main paper. The main article retains the results needed to establish the historical reference effect, evaluate its evolution over calendar time, and interpret contemporary attenuation. The Supplement gives the technical detail needed to assess effect-scale choices, small-sample inference, heterogeneity, prediction, historical completeness, modern sampling variances, temporal-model sensitivity, and explanatory moderator analyses.

S1 Frequentist Benchmark for the Nine Historical Trials

This section documents the reproducible frequentist benchmark used for the nine-study historical evidence set. Its role is deliberately limited: the primary inferential framework in the main paper is Bayesian, while the REML–Hartung–Knapp analysis provides a classical reference for the pooled historical effect, heterogeneity, prediction, and study influence. All calculations reported here were regenerated directly from the arm-level means, standard deviations, and sample sizes rather than from rounded effect estimates printed in the manuscript.

S1.1 Arm-level historical input data

For reproducibility, Table S1 records the arm-level summaries from which the nine historical log ratios are regenerated. These are the frozen inputs used by the production scripts; study-level effects are not entered independently as rounded manuscript values.

The trials are not statistically interchangeable in design. In particular, Milsom is school based, while several older reports predate current standards for reporting design-adjusted uncertainty. The arm summaries above define the historical benchmark; separate sensitivity analyses below address effect-scale approximation, study influence, and historical completeness.

Table S1: Arm-level summary data for the nine historical trials.

Study	Fluoride varnish			Control		
	Mean	SD	n	Mean	SD	n
Koch	0.90	3.80	60	4.00	3.75	61
Modeer	2.50	3.10	87	3.70	3.90	107
Clark	2.43	3.09	246	3.11	3.54	234
Tewari	0.55	4.48	311	2.16	4.12	307
Bravo	1.48	1.53	98	2.58	1.89	116
Sköld	0.79	1.67	190	1.85	2.89	181
Arruda	4.61	4.39	113	7.72	5.84	97
Tagliaferro	0.33	1.04	91	0.57	1.39	86
Milsom	0.66	0.73	94	0.63	0.66	95

S1.2 Study-level effect calculation

For study i , the observed effect is the log ratio of the treatment and control means,

$$y_i = \log \left(\frac{\bar{X}_{Ti}}{\bar{X}_{Ci}} \right),$$

with first-order delta-method sampling variance

$$v_i = \text{Var}(y_i) \approx \frac{s_{Ti}^2}{n_{Ti}\bar{X}_{Ti}^2} + \frac{s_{Ci}^2}{n_{Ci}\bar{X}_{Ci}^2}.$$

The corresponding positive prevented fraction is

$$\text{PF}_{+,i} = 1 - \exp(y_i).$$

Because this transformation is decreasing in y_i , the endpoints of an interval on the log-ratio scale reverse when transformed to the prevented-fraction scale.

The primary analysis uses the ordinary first-order log response ratio. This point is made explicit for reproducibility because current versions of `metafor` can apply a second-order bias correction to the log ratio of means. The production calculation therefore uses `measure="ROM"`, `vtype="LS"`, and `correct=FALSE`. A second-order bias-corrected response-ratio analysis is treated separately as a sensitivity analysis, particularly because Koch and Tewari have treatment means that are small relative to their reported variability.[2, 18]

Table S2 gives the study-level quantities reproduced by the production script.

S1.3 REML estimation and Hartung–Knapp inference

The frequentist benchmark uses the random-effects model

$$y_i \mid \theta_i \sim N(\theta_i, v_i), \quad \theta_i \mid \mu, \tau \sim N(\mu, \tau^2).$$

The heterogeneity variance τ^2 is estimated by restricted maximum likelihood. For inference on the pooled mean, the Hartung–Knapp adjustment is used rather than a normal/Wald approximation.[27]

Table S2: Study-level log ratios and first-order delta-method standard errors for the nine historical trials. The prevented fraction is $\text{PF}_+ = 1 - \exp(y)$.

Study	y_i	$\text{SE}(y_i)$	$\widehat{\text{PF}}_{+,i}$	95% interval for $\text{PF}_{+,i}$
Koch	-1.4917	0.5581	0.7750	[0.3281, 0.9246]
Modeer	-0.3920	0.1675	0.3243	[0.0618, 0.5134]
Clark	-0.2467	0.1100	0.2186	[0.0306, 0.3702]
Tewari	-1.3679	0.4745	0.7454	[0.3546, 0.8995]
Bravo	-0.5557	0.1246	0.4264	[0.2676, 0.5507]
Sköld	-0.8509	0.1924	0.5730	[0.3774, 0.7071]
Arruda	-0.5156	0.1180	0.4028	[0.2475, 0.5262]
Tagliaferro	-0.5465	0.4222	0.4211	[-0.3245, 0.7469]
Milsom	0.0465	0.1567	-0.0476	[-0.4244, 0.2295]

For an intercept-only random-effects model, define

$$w_i = \frac{1}{v_i + \widehat{\tau}^2},$$

and

$$q_{\text{HK}} = \frac{1}{k-1} \sum_{i=1}^k w_i (y_i - \widehat{\mu})^2.$$

The Hartung–Knapp standard error can then be written as

$$\text{SE}_{\text{HK}}(\widehat{\mu}) = \left(\frac{q_{\text{HK}}}{\sum_i w_i} \right)^{1/2}.$$

For the present nine studies,

$$q_{\text{HK}} = 1.2139 > 1.$$

Consequently, the ordinary Hartung–Knapp procedure and the modified Hartung–Knapp safeguard coincide; no artificial narrowing of the interval occurs.

The production REML fit gives

$$\widehat{\mu} = -0.505028, \quad \widehat{\tau}^2 = 0.080399, \quad \widehat{\tau} = 0.283548,$$

with

$$\text{SE}_{\text{HK}}(\widehat{\mu}) = 0.132839.$$

The 95% Hartung–Knapp interval on the log-ratio scale is

$$[-0.811356, -0.198701].$$

After transformation,

$$\widehat{\text{PF}}_{+,\text{pool}} = 1 - \exp(-0.505028) = 0.396512,$$

with 95% confidence interval

$$[0.180205, 0.555745].$$

The heterogeneity diagnostics also indicate substantial cross-study variation:

$$Q = 25.9883, \quad df = 8, \quad p = 0.00106,$$

and

$$I^2 = 73.52\%, \quad H^2 = 3.776.$$

These quantities are descriptive summaries of heterogeneity; the model itself is parameterized through τ .

S1.4 Prediction for a new true study effect

Prediction is reported because uncertainty about transportability is materially larger than uncertainty about the pooled historical mean. We use the small-sample prediction interval

$$\hat{\mu} \pm t_{k-2, 0.975} \left\{ \hat{\tau}^2 + \text{SE}_{\text{HK}}(\hat{\mu})^2 \right\}^{1/2},$$

which uses a t_{k-2} reference distribution.[?, 16] For $k = 9$, the resulting 95% prediction interval on the log-ratio scale is

$$[-1.245445, 0.235388].$$

Transforming to the prevented-fraction scale gives

$$[-0.265399, 0.712187].$$

Thus the pooled historical mean is clearly preventive, while the interval for a new true study effect is substantially wider and includes effects near the null and on the adverse side.

S1.5 Second-order log-response-ratio sensitivity

Because the first-order log-response-ratio approximation can be less accurate when an arm mean is small relative to its variability, we also apply the second-order correction of Lajeunesse.[18] The corrected effect and sampling variance are

$$y_i^{(2)} = y_i + \frac{1}{2} \left[\frac{s_{Ti}^2}{n_{Ti} \bar{X}_{Ti}^2} - \frac{s_{Ci}^2}{n_{Ci} \bar{X}_{Ci}^2} \right], \quad (\text{S1})$$

and

$$v_i^{(2)} = \frac{s_{Ti}^2}{n_{Ti} \bar{X}_{Ti}^2} + \frac{s_{Ci}^2}{n_{Ci} \bar{X}_{Ci}^2} + \frac{s_{Ti}^4}{2n_{Ti}^2 \bar{X}_{Ti}^4} + \frac{s_{Ci}^4}{2n_{Ci}^2 \bar{X}_{Ci}^4}. \quad (\text{S2})$$

The largest corrections occur for Koch and Tewari. Re-fitting the REML–Hartung–Knapp model to the corrected quantities gives a pooled log ratio of approximately -0.478 , corresponding to $\widehat{\text{PF}}_+ = 0.380$ with 95% confidence interval $[0.183, 0.529]$. The corrected heterogeneity estimate is $\hat{\tau} = 0.256$, and the 95% prediction interval is approximately $[-0.210, 0.682]$ on the positive prevented-fraction scale. The correction therefore changes the numerical center only modestly and does not alter the historical interpretation.

S1.6 Leave-one-study-out influence analysis

The full random-effects analysis was repeated nine times, deleting one study at a time. Table S4 reports the resulting pooled prevented fraction, Hartung–Knapp confidence interval, REML heterogeneity estimate, and small-sample prediction interval.

Table S3: First-order and second-order log response ratios for the nine historical studies.

Study	First-order LRR	Corrected LRR	Corrected SE
Koch	-1.4917	-1.3503	0.5965
Modeer	-0.3920	-0.3884	0.1681
Clark	-0.2467	-0.2462	0.1102
Tewari	-1.3679	-1.2672	0.4980
Bravo	-0.5557	-0.5526	0.1249
Sköld	-0.8509	-0.8459	0.1933
Arruda	-0.5156	-0.5145	0.1182
Tagliaferro	-0.5465	-0.5265	0.4320
Milsom	0.0465	0.0473	0.1572

Table S4: Leave-one-study-out REML–Hartung–Knapp analysis for the nine historical trials. All effect summaries are shown on the positive prevented-fraction scale.

Omitted study	Pooled PF ₊	95% CI	$\hat{\tau}$	95% PI
Koch	0.369	[0.159, 0.527]	0.256	[-0.263, 0.685]
Modeer	0.418	[0.157, 0.598]	0.332	[-0.431, 0.763]
Clark	0.430	[0.187, 0.601]	0.310	[-0.325, 0.755]
Tewari	0.365	[0.153, 0.524]	0.250	[-0.254, 0.679]
Bravo	0.403	[0.130, 0.590]	0.334	[-0.476, 0.758]
Sköld	0.357	[0.112, 0.535]	0.246	[-0.280, 0.677]
Arruda	0.408	[0.136, 0.594]	0.338	[-0.477, 0.763]
Tagliaferro	0.399	[0.146, 0.578]	0.305	[-0.376, 0.738]
Milsom	0.425	[0.254, 0.556]	0.183	[0.030, 0.659]

The pooled center is stable: deleting any one study gives an estimated prevented fraction between 0.357 and 0.430. Heterogeneity and prediction are more sensitive. The most consequential deletion is Milsom, for which $\hat{\tau}$ falls from 0.284 to 0.183 and the entire 95% prediction interval becomes preventive. Milsom therefore affects the estimated transportability of the historical effect more strongly than it affects the pooled average.

S1.7 Funnel-plot interpretation and small-study effects

With only nine historical studies and substantial between-study heterogeneity, funnel-plot asymmetry is difficult to distinguish from genuine effect modification, differences in design, or chance. Any funnel plots are therefore treated as descriptive diagnostics rather than as evidence for or against publication bias. We do not use a formal asymmetry test as a primary inferential result because such tests have low power with so few studies and can be misleading when heterogeneity is large.

S1.8 Software implementation and reproducibility

The production benchmark was run under R version 4.5.3 on Windows 11 using `metafor` version 5.2-1. The arm-level data were entered once and all study-level effects, sampling variances, pooled estimates, confidence intervals, prediction intervals, influence analyses, and figures were generated from that single data object. The core implementation is

```

dat_es <- escalc(
  measure = "ROM",
  m1i = mean_T, sd1i = sd_T, n1i = n_T,
  m2i = mean_C, sd2i = sd_C, n2i = n_C,
  data = dat,
  vtype = "LS",
  correct = FALSE
)

fit_hk <- rma.uni(
  yi = yi,
  vi = vi,
  data = dat,
  method = "REML",
  test = "knha",
  level = 95
)

pred_hk <- predict(
  fit_hk,
  level = 95,
  predtype = "Riley"
)

```

The production script independently computes the log ratios and delta-method variances from the variance formula above and verifies numerically that they agree with the `metafor` calculations. It also checks the Hartung–Knapp scale factor explicitly and reconstructs the small-sample prediction interval in the small-sample prediction formula above independently of the package output. This redundancy is intentional: it guards against silent changes in software defaults and makes the numerical benchmark reproducible from the reported arm summaries.

The archived output consists of the study-level effect table, the REML–HK summary, the complete leave-one-out table, a forest plot, influence diagnostics, and the R session information. The same frozen input dataset and software record should be retained with the final manuscript.

S2 Historical Completeness Sensitivity Analysis: the Cochrane-13 Evidence Set

The principal historical analysis in the main paper is based on the defined nine-study arm-level dataset and uses the log ratio of means as its preferred effect scale. The purpose of the present analysis is different. It asks whether the historical pooled effect is materially changed when the nine-study subset is expanded to the complete set of 13 trials contributing to the permanent-tooth-surface $D(M)FS$ prevented-fraction analysis in the 2013 Cochrane review.[21] This is therefore a *historical-completeness sensitivity analysis*, not an alternative primary likelihood.

A common historical scale is essential for this comparison. Consequently, the nine-study subset in this section is represented by the prevented-fraction estimates and standard errors reported in the same Cochrane forest plot used for the 13-study analysis. These values are not substituted by the arm-level log-ratio quantities used in the principal analysis. This distinction avoids comparing nine studies on one effect construction with thirteen studies on another.

S2.1 Cochrane-scale study-level data

Let z_i denote the conventional positive prevented fraction for study i and s_i its synthesis-level standard error. Positive values favor fluoride varnish. The 13 study-level pairs used in the reconstruction are shown in Table S5. The four studies not contained in the nine-study historical core are Holm (1984), Borutta (1991), Hardman (2007), and Gugwad (2011).[14, 3, 12, 11]

Table S5: Study-level prevented fractions and standard errors used in the Cochrane-13 historical-completeness sensitivity analysis. An asterisk denotes one of the four trials added to the original nine-study subset.

Study	Prevented fraction	SE
Koch 1975	0.78	0.126
Modeer 1984	0.30	0.154
Holm 1984*	0.55	0.078
Clark 1985	0.20	0.077
Tewari 1990	0.75	0.124
Borutta 1991*	0.29	0.106
Bravo 1997	0.43	0.141
Sköld 2005	0.60	0.060
Hardman 2007*	-0.18	0.451
Tagliaferro 2011	0.42	0.245
Gugwad 2011*	1.88	0.832
Milsom 2011	0.03	0.144
Arruda 2012	0.40	0.091

The use of the Cochrane-scale quantities is particularly important for the four added studies because each raises a different reconstruction or design issue. These issues are documented explicitly below.

S2.2 Study-specific considerations for the four additional trials

Holm (1984). Holm et al. studied newly erupted first permanent molars over approximately 24 months with repeated Duraphat applications.[14] The principal issue is reconstruction of the variance corresponding to the caries-increment quantity used in the synthesis. The primary report emphasizes fissure- or tooth-surface-level outcomes, and the exact mapping from the published summaries to the synthesis-level increment variance is less transparent than for the nine core studies. We therefore retain the Cochrane-scale effect and uncertainty rather than constructing a new arm-level variance from incomplete historical summaries.

Borutta (1991). Borutta et al. used a multi-arm, placebo-controlled design with three active varnish regimens sharing one placebo group.[3] The resulting active-versus-placebo contrasts are statistically dependent. Treating them as three independent studies would reuse the same control information and artificially increase precision. The historical-completeness analysis therefore retains a single study-level contrast corresponding to the Cochrane synthesis. A direct arm-level reanalysis would require a multivariate likelihood that explicitly shares the placebo arm.

Hardman (2007). Hardman et al. conducted a school-based cluster-randomized trial involving 24 primary schools.[12] Children within the same school cannot be treated as statistically independent, so a standard error reconstructed from nominal child counts would generally be too small. We therefore retain the synthesis-level effect and uncertainty rather than imposing a naive child-level variance. The study’s prevented-fraction estimate is also imprecise, approximately -0.18 with SE 0.451 , and consequently receives relatively little weight.

Gugwad (2011). Gugwad et al. contributes a reconstructed prevented fraction of approximately 1.88 with SE 0.832 . [11] This is not merely an extreme value: it lies outside the support of the log-ratio-derived quantity

$$\text{PF}_+ = 1 - \exp(\theta) < 1$$

for positive arm means. Forcing Gugwad into the preferred log-ratio likelihood would therefore require an artificial transformation or an unverifiable reconstruction. It is retained on the legacy prevented-fraction scale where the historical synthesis naturally represents it.

Together these cases explain why the 13-study analysis is a completeness sensitivity analysis rather than a replacement for the preferred nine-study log-ratio likelihood.

S2.3 Common random-effects model

For this comparison only, both evidence sets are analyzed using the same normal–normal model on the prevented-fraction scale,

$$z_i \mid \phi_i \sim N(\phi_i, s_i^2), \quad \phi_i \mid \mu, \tau \sim N(\mu, \tau^2).$$

The model is intentionally simple. The prevented fraction is not naturally Gaussian over the entire real line, and this Gaussian PF-scale model is not used for the primary inference in the paper. Its purpose is to hold the effect scale and likelihood fixed while changing only the historical evidence set from nine to thirteen studies.

The heterogeneity variance is estimated by restricted maximum likelihood. Hartung–Knapp inference is computed as a small-sample benchmark, together with the modified Hartung–Knapp safeguard.[27] Prediction for a new true study effect is based on the same t_{k-2} small-sample formulation used elsewhere in the frequentist analyses.[?, 16]

S2.4 Nine versus thirteen studies

For the nine-study Cochrane-scale subset, the REML fit gives

$$\hat{\mu}_9 = 0.4394, \quad \hat{\tau}_9 = 0.2169.$$

For the full 13-study set,

$$\hat{\mu}_{13} = 0.4346, \quad \hat{\tau}_{13} = 0.1974.$$

The change in pooled prevented fraction is therefore

$$\hat{\mu}_{13} - \hat{\mu}_9 = -0.0048,$$

or less than one-half of one percentage point. The estimated heterogeneity standard deviation also decreases slightly after adding the four trials.

Table S6 gives the corresponding uncertainty summaries. For the nine-study PF-scale fit, the ordinary Hartung–Knapp scale factor is slightly below one, so the modified Hartung–Knapp safeguard gives a marginally wider confidence interval. For the 13-study fit, the ordinary and modified Hartung–Knapp intervals coincide.

Table S6: Historical-completeness sensitivity analysis on the common Cochrane prevented-fraction scale. Frequentist intervals use the modified Hartung–Knapp safeguard and the small-sample prediction formulation. Bayesian summaries use a diffuse prior for the mean and $\tau \sim \text{Half-Normal}(0, 0.5^2)$.

Quantity	Nine-study subset	Cochrane-13
Number of studies	9	13
REML pooled PF, $\hat{\mu}$	0.439	0.435
REML heterogeneity SD, $\hat{\tau}$	0.217	0.197
95% modified-HK CI for μ	[0.246, 0.633]	[0.272, 0.597]
95% frequentist prediction interval	[−0.111, 0.989]	[−0.030, 0.899]
Bayesian posterior median PF	0.436	0.433
Bayesian posterior median τ	0.238	0.214
95% Bayesian predictive interval	[−0.15, 1.01]	[−0.08, 0.94]

The pooled historical center is therefore nearly invariant to expansion from nine to thirteen trials under both inferential frameworks. The main effect of adding the four studies is a modest reduction in estimated heterogeneity and a somewhat narrower prediction interval. This is precisely the role of the Cochrane-13 analysis in the paper: it provides evidence that the historical center inferred from the nine-study core is not an artifact of omitting those four permanent-surface trials.

S2.5 Relation to the published Cochrane synthesis

The 2013 Cochrane review reported an overall prevented fraction of approximately 43% for the 13 permanent-surface trials, with a 95% confidence interval of approximately 30% to 57% and substantial heterogeneity (I^2 about 75%).^[21] The present REML reconstruction is not intended to reproduce every detail of the original Review Manager estimator. Rather, it uses the study-level prevented fractions and standard errors from the Cochrane forest plot and reanalyzes them under the same REML framework used for the present paper. The resulting pooled center, 0.435, is essentially identical to the published historical summary.

The four additional studies should therefore not be interpreted as creating a new evidence base or changing the scientific conclusion. Their function is to test historical data completeness. Because their designs and reconstructed effect summaries are not fully compatible with the preferred modern log-ratio-of-means likelihood, the Cochrane-13 analysis remains supplementary.

S2.6 Software implementation and reproducibility

The analysis is implemented in R using the `metafor` package. The core computations are

```
fit9 <- rma.uni(
  yi = PF, sei = SE,
```

```

  data = dat[dat$core9, ],
  method = "REML",
  test = "adhoc"
)

fit13 <- rma.uni(
  yi = PF, sei = SE,
  data = dat,
  method = "REML",
  test = "adhoc"
)

pred9 <- predict(fit9, predtype = "Riley")
pred13 <- predict(fit13, predtype = "Riley")

```

The production script stores the 13 Cochrane-scale study-level effects in one data frame, identifies the nine-study subset with a logical indicator, fits both models from the same object, checks the reproduced values against the rounded manuscript targets, generates the 13-study forest plot and a direct nine-versus-thirteen summary figure, and archives the numerical results and `sessionInfo()`. This structure makes the completeness claim fully reproducible while keeping the analysis distinct from the primary log-ratio model.

S3 Bayesian Normal–Normal Hierarchy and Prior Sensitivity

The principal Bayesian analysis of the nine historical trials uses the same study-specific log ratios and first-order sampling standard errors as the frequentist REML–Hartung–Knapp benchmark. The purpose of this section is to document the Bayesian computation in full and to determine how much the historical pooled effect, heterogeneity, and prediction depend on the prior assigned to the between-study standard deviation.

Unlike the aggregate arm-level and heavy-tailed models considered later, the normal–normal hierarchical model does not require Markov chain Monte Carlo. The `bayesmeta` package exploits the conditional Gaussian structure of the model and evaluates the posterior by deterministic numerical integration.[25] Consequently, MCMC diagnostics such as $\hat{R}$, effective sample sizes, Monte Carlo standard errors, divergences, and tree-depth diagnostics are not applicable to this part of the analysis. Those diagnostics are reserved for the Stan models described elsewhere in the Supplement.

S3.1 Model and common prior for the pooled effect

For study $i = 1, \dots, 9$, let

$$y_i = \log \left(\frac{\bar{X}_{Ti}}{\bar{X}_{Ci}} \right)$$

and let s_i denote its first-order delta-method standard error. The model is

$$y_i \mid \theta_i \sim N(\theta_i, s_i^2), \quad \theta_i \mid \mu, \tau \sim N(\mu, \tau^2).$$

The pooled log-ratio parameter is assigned the common prior

$$\mu \sim N(0, 1^2)$$

in every sensitivity analysis. This prior is deliberately broad on the multiplicative effect scale: a 95% prior interval for μ of approximately $[-1.96, 1.96]$ corresponds to treatment-to-control mean ratios from roughly 0.14 to 7.1. The prior is therefore not intended to provide substantive shrinkage of the pooled treatment effect.

The quantity reported clinically is

$$\text{PF}_{+, \text{pool}} = 1 - \exp(\mu).$$

Prediction concerns a new *true* study effect,

$$\theta_{\text{new}} \mid \mu, \tau \sim N(\mu, \tau^2),$$

with

$$\text{PF}_{+, \text{new}} = 1 - \exp(\theta_{\text{new}}).$$

Thus

$$\Pr(\text{PF}_{+, \text{new}} > 0 \mid \mathcal{D}) = \Pr(\theta_{\text{new}} < 0 \mid \mathcal{D}),$$

and

$$\Pr(\text{PF}_{+, \text{new}} > 0.20 \mid \mathcal{D}) = \Pr\{\theta_{\text{new}} < \log(0.80) \mid \mathcal{D}\}.$$

S3.2 Heterogeneity priors

Because only nine studies are available, the likelihood is considerably less informative about τ than about μ . We therefore use a small set of interpretable priors that vary in both concentration and tail behavior rather than a large collection of nominally noninformative variance priors. [9, 26, 20]

The primary specification is

$$\tau \sim \text{Half-Normal}(0, 0.5^2).$$

As an effect-scale calibration, $\tau = 0.5$ implies that approximately 95% of normally distributed true study effects lie within $\mu \pm 0.98$, corresponding to treatment-to-control mean ratios differing from the pooled ratio by a factor of approximately

$$\exp(1.96 \times 0.5) \approx 2.66.$$

The prior therefore allows substantial heterogeneity while progressively downweighting very large values.

A more concentrated alternative is

$$\tau \sim \text{Half-Normal}(0, 0.25^2),$$

for which $\tau = 0.25$ corresponds to a multiplicative factor of approximately

$$\exp(1.96 \times 0.25) \approx 1.63.$$

To examine sensitivity to heavier tails while retaining a comparable central scale, we also use

$$\tau \sim \text{Half-}t_4(0, 0.5).$$

This prior assigns more probability to large heterogeneity than the primary half-normal prior and therefore provides a direct check on whether the predictive conclusions are being driven by Gaussian prior tails.

Finally, for continuity with the earlier analysis we use the proper DuMouchel prior

$$\pi(\tau) = \frac{s_0}{(s_0 + \tau)^2}, \quad \tau \geq 0,$$

where

$$s_0^2 = \frac{k}{\sum_{i=1}^k s_i^{-2}}.$$

For the present log-ratio standard errors,

$$s_0 \approx 0.164.$$

The DuMouchel prior therefore has substantial mass near zero but a much heavier right tail than either half-normal prior. It constitutes a useful shape-sensitivity analysis rather than a preferred default.[6]

S3.3 Deterministic posterior computation

Conditional on τ , the normal-normal hierarchy above is Gaussian in μ . The `bayesmeta` calculation integrates over τ numerically and thereby obtains the marginal posteriors of μ and τ , the study-specific shrinkage distributions, and the posterior predictive distribution without simulation error from an MCMC chain.[25]

The production calculation uses central equal-tailed 95% intervals and tighter numerical integration tolerances than the package defaults. The core implementation is

```
fit <- bayesmeta(
  y = dat$yi,
  sigma = dat$sei,
  labels = dat$study,
  mu.prior.mean = 0,
  mu.prior.sd = 1,
  tau.prior = tau_prior,
  interval.type = "central",
  delta = 0.005,
  epsilon = 1e-6
)
```

The primary half-normal, concentrated half-normal, and half- t_4 priors are passed as explicit density functions. The DuMouchel prior is invoked through `tau.prior="DuMouchel"`, which uses the within-study standard errors to determine the scale s_0 in the DuMouchel prior above.

S3.4 Posterior sensitivity to the heterogeneity prior

Table S7 summarizes the complete prior-sensitivity analysis. Posterior medians are reported because the transformation $PF_+ = 1 - \exp(\cdot)$ is nonlinear; central 95% intervals are obtained on the log-ratio scale and transformed with the endpoints reversed.

Table S7: Sensitivity of the nine-study Bayesian normal-normal hierarchical model to the prior for the heterogeneity standard deviation. Entries are posterior medians; intervals are central 95% credible or predictive intervals.

Prior for τ	Pooled PF ₊	τ	Predictive PF _{+,new}	Pr(PF _{+,new} > 0)	Pr(PF _{+,new} > 0.20)
Half-N(0, 0.5 ²)	0.389 [0.215, 0.556]	0.302 [0.100, 0.661]	0.386 [−0.284, 0.732]	0.927	0.809
Half-N(0, 0.25 ²)	0.382 [0.233, 0.529]	0.253 [0.077, 0.501]	0.379 [−0.127, 0.683]	0.951	0.838
Half- t_4 (0, 0.5)	0.389 [0.215, 0.556]	0.300 [0.098, 0.670]	0.386 [−0.284, 0.732]	0.927	0.809
DuMouchel	0.380 [0.229, 0.538]	0.249 [0.044, 0.604]	0.375 [−0.170, 0.700]	0.947	0.839

The main conclusion is that the posterior center is remarkably stable. Across all four heterogeneity priors, the posterior median pooled prevented fraction lies between 0.380 and 0.389, a spread of less than one percentage point. The 95% credible intervals for the pooled effect also overlap almost completely. The historical average treatment effect is therefore being determined primarily by the likelihood rather than by fine distinctions among these defensible heterogeneity priors.

The consequences of the prior are more visible for τ and for prediction. The more concentrated half-normal prior lowers the posterior median heterogeneity from about 0.302 to 0.253 and narrows the 95% predictive interval from approximately [−0.284, 0.732] to [−0.127, 0.683]. The probability that a new comparable study has a preventive true effect rises from approximately 0.927 to 0.951. The DuMouchel prior gives a similar posterior median for τ but retains appreciable right-tail uncertainty; its resulting predictive distribution lies between those produced by the concentrated and primary half-normal specifications.

The half- t_4 sensitivity produces a posterior nearly indistinguishable from the primary half-normal analysis in the region supported by the data. Its heavier prior tail is visible mainly in the upper tail of τ , not in the posterior pooled center. This comparison is useful because it shows that the primary result is not an artifact of the exponentially decreasing tail of the half-normal prior.

Figure S1 displays the four prior densities, and Figure S2 shows the resulting marginal posteriors for τ .

S3.5 Pooled and predictive distributions

The marginal posterior distributions of the pooled log ratio are much less sensitive to the heterogeneity prior than the posterior distributions of τ . Figure S3 shows that the four marginal posteriors for μ nearly superimpose.

Prediction shows the opposite pattern. Because $\theta_{\text{new}} \mid \mu, \tau \sim N(\mu, \tau^2)$, uncertainty in τ enters directly into the posterior predictive tails. Figure S4 therefore separates the prior specifications more clearly than the corresponding pooled-effect plot.

This distinction is substantively important. The data strongly constrain the historical average, but they provide less information about the dispersion of true effects across settings. Consequently, reasonable prior choices matter more for statements about transportability than for the pooled historical effect.

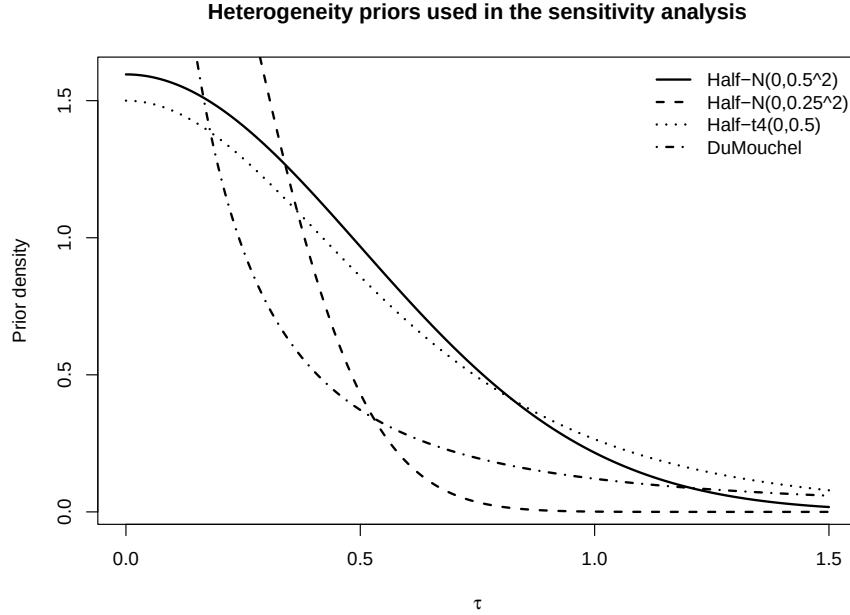


Figure S1: Prior densities for the heterogeneity standard deviation τ in the nine-study Bayesian sensitivity analysis. The DuMouchel prior is scaled using the harmonic-mean within-study variance implied by the nine log-ratio standard errors.

S3.6 Study-specific shrinkage under the primary prior

The production script also evaluates the posterior distribution of each study-specific true effect under the primary Half-Normal($0, 0.5^2$) heterogeneity prior. These distributions combine the study-specific estimate with information from the remaining trials. Imprecise extreme estimates are pulled more strongly toward the pooled effect, whereas precise mid-range studies undergo less shrinkage.

The complete numerical shrinkage table is written by the production script to `bayesmeta_primary_shrinkage.csv`. Figure S5 compares the raw study prevented fractions with the posterior medians and 95% credible intervals.

S3.7 Reproducibility

The production script reconstructs the nine log ratios and first-order delta-method standard errors directly from the frozen arm-level dataset, fits all four prior specifications from those same inputs, and writes the full posterior summaries, paper-level table, prior calibration, study-specific shrinkage results, and session information to a single output directory. The script also performs a numerical benchmark check against independently computed deterministic posterior summaries. This check is intended to catch errors in effect-scale definition, prior parameterization, or transformation to the prevented-fraction scale.

No random-number seed is needed for the `bayesmeta` calculations reported in this section because the posterior summaries are obtained by deterministic integration rather than Monte Carlo simulation. The archived R and `bayesmeta` version information should nevertheless be reported with the final

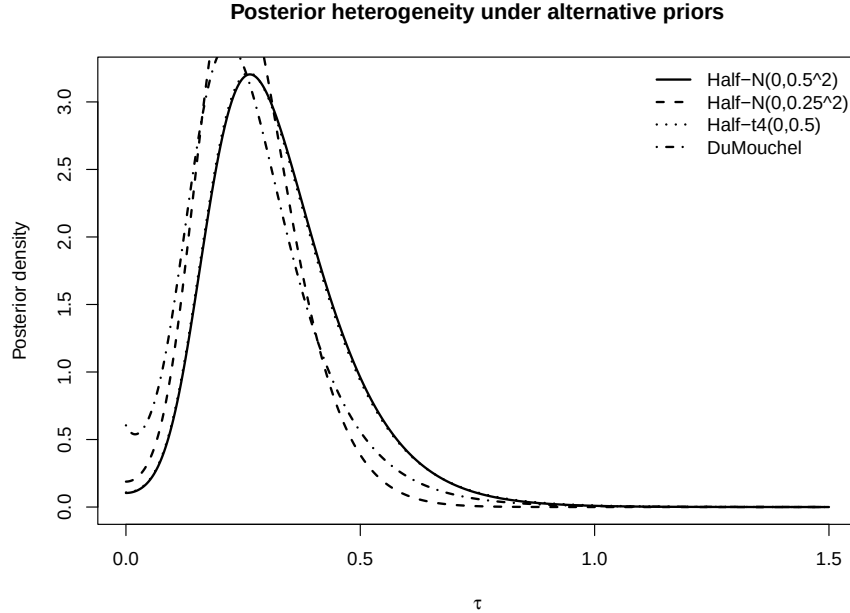


Figure S2: Marginal posterior densities of τ under the four heterogeneity priors. The posterior is more concentrated than the prior in each case, but the upper tail remains prior-sensitive because only nine studies are available.

analysis because numerical defaults and package implementations can change over time.

S4 Stan Production Analysis of the Aggregate Arm-Level Models

The aggregate arm-level Gaussian and robust t_4 random-effects models were regenerated in Stan from the frozen nine-study arm summaries. These analyses complement the contrast-based normal-normal hierarchical model by avoiding a first-stage Gaussian approximation to the estimated log ratio of means. Instead, the reported treatment and control sample means enter the likelihood directly. The resulting likelihood is nonlinear in the latent baseline means and study-specific treatment effects, making Hamiltonian Monte Carlo a natural computational approach.[4]

S4.1 Aggregate arm-level likelihood

For study $i = 1, \dots, 9$, let $m_i > 0$ denote the latent control-arm mean and let θ_i denote the study-specific log treatment-to-control mean ratio. The sampling model is

$$\bar{X}_{Ci} \mid m_i \sim N\left(m_i, \frac{s_{Ci}^2}{n_{Ci}}\right),$$

and

$$\bar{X}_{Ti} \mid m_i, \theta_i \sim N\left(m_i e^{\theta_i}, \frac{s_{Ti}^2}{n_{Ti}}\right).$$

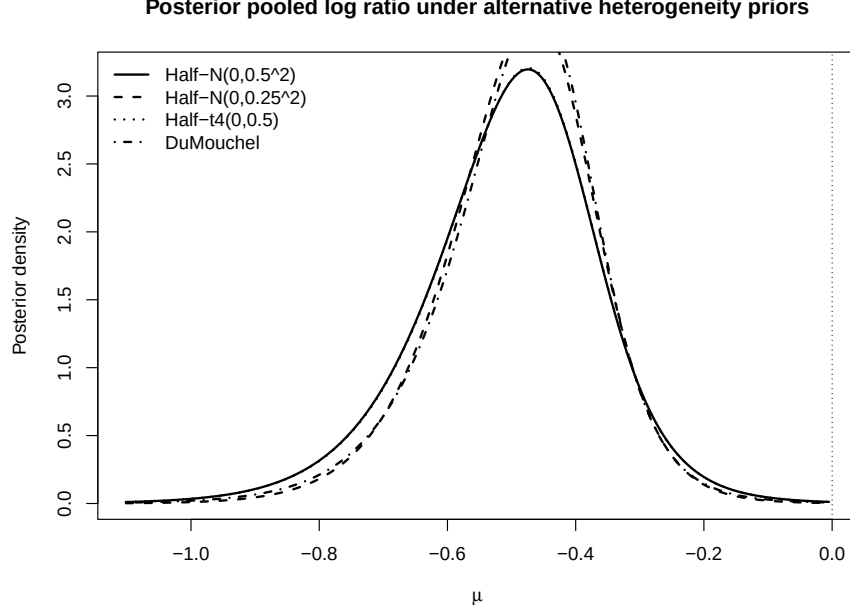


Figure S3: Marginal posterior densities of the pooled log ratio μ under the four heterogeneity priors. The near overlap of the curves illustrates the stability of the historical pooled center.

Positivity of the latent baseline mean is imposed through

$$m_i = \exp(\eta_i), \quad \eta_i \sim N(0, 2.5^2).$$

The common priors are

$$\mu \sim N(0, 1^2), \quad \tau \sim \text{Half-Normal}(0, 0.5^2).$$

The reported arm standard deviations and sample sizes are treated as fixed. Consequently, this remains an aggregate-data approximation. It does not reconstruct zero inflation, within-person longitudinal dependence, cluster effects not represented in the reported uncertainty, or individual-level covariates.

S4.2 Gaussian and standardized t_4 random effects

For the Gaussian model, the random effects were parameterized noncentrally as

$$\theta_i = \mu + \tau z_i, \quad z_i \sim N(0, 1).$$

The robust model retains the same definitions of μ and τ but replaces the Gaussian standardized effect by

$$\theta_i = \mu + \tau z_i, \quad z_i \sim t_4 \left(0, \sqrt{\frac{4-2}{4}} \right).$$

For a Student- t_ν distribution with scale s ,

$$\text{Var}(Z) = s^2 \frac{\nu}{\nu - 2}, \quad \nu > 2.$$

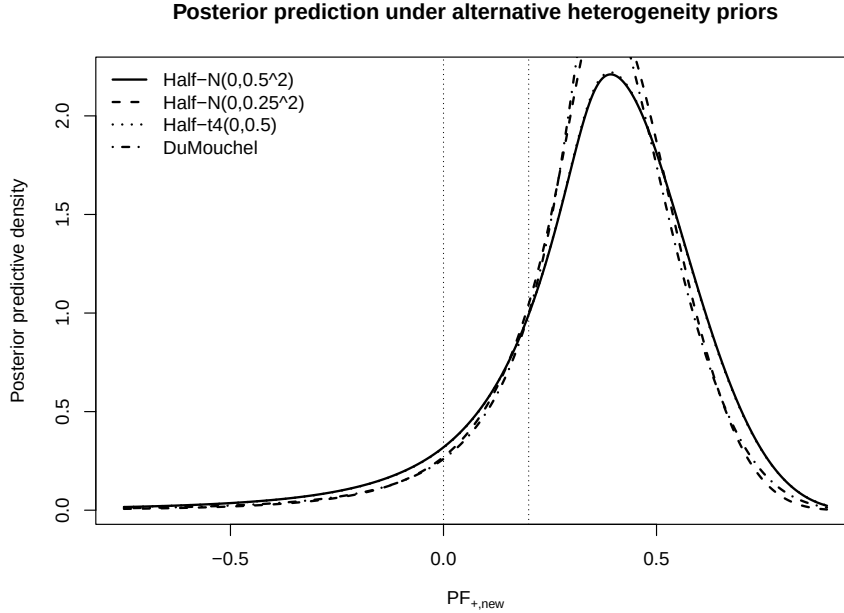


Figure S4: Posterior predictive densities for the true prevented fraction in a new comparable study under the four heterogeneity priors. Vertical reference lines mark no preventive effect and a 20% preventive fraction.

Thus the scale $s = \sqrt{(4 - 2)/4} = 1/\sqrt{2}$ gives $\text{Var}(z_i) = 1$, so that τ retains the interpretation of a between-study standard deviation under both random-effects distributions. The degrees of freedom were fixed at four rather than estimated because nine studies provide little information for learning a separate tail parameter.

S4.3 Posterior targets and prediction

For each posterior draw, the pooled prevented fraction was calculated as

$$\text{PF}_{+, \text{pool}} = 1 - e^{\mu}.$$

A new true study effect was then generated from the corresponding fitted random-effects distribution,

$$\theta_{\text{new}} \mid \mu, \tau,$$

and transformed to

$$\text{PF}_{+, \text{new}} = 1 - e^{\theta_{\text{new}}}.$$

We also estimated

$$\Pr(\text{PF}_{+, \text{new}} > 0 \mid \mathcal{D})$$

and

$$\Pr(\text{PF}_{+, \text{new}} > 0.20 \mid \mathcal{D}),$$

which quantify, respectively, the posterior probability of any preventive effect and of at least a 20% preventive fraction in a new comparable study.

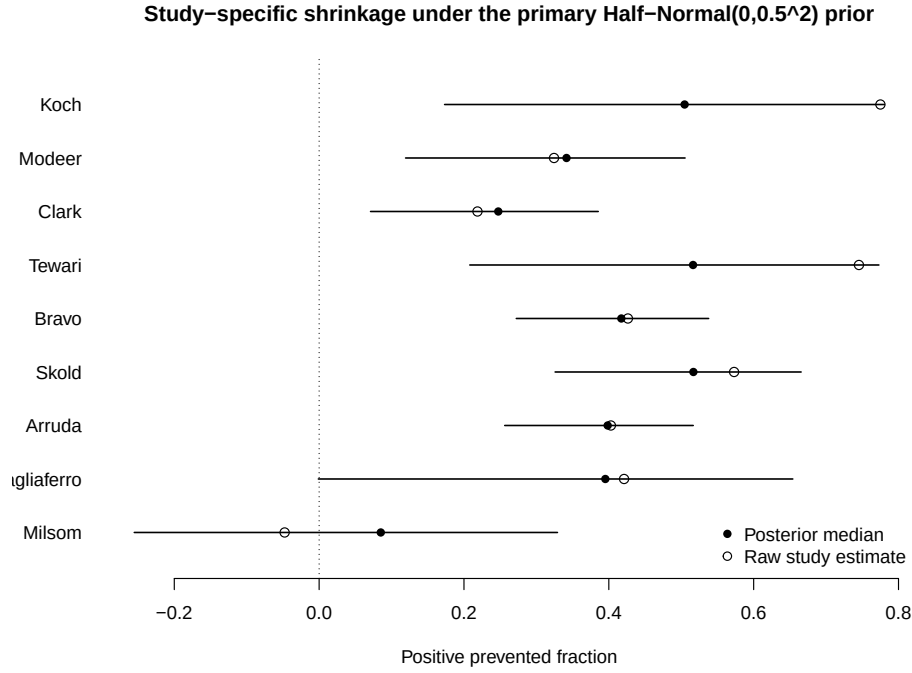


Figure S5: Study-specific posterior shrinkage under the primary Half-Normal($0, 0.5^2$) heterogeneity prior. Filled points show posterior medians and horizontal segments show central 95% credible intervals; open points denote the corresponding raw study prevented-fraction estimates.

S4.4 NUTS implementation and convergence assessment

Each model was fit with four independently initialized NUTS chains, 2000 warm-up iterations per chain, and 4000 retained iterations per chain, giving 16,000 post-warm-up draws per model. The target acceptance probability was 0.995, the maximum tree depth was 15, and the random effects were sampled using the noncentered parameterization above.

Convergence and Hamiltonian Monte Carlo geometry were assessed using rank-normalized split $\hat{R}$, bulk and tail effective sample sizes, Monte Carlo standard errors, divergent transitions, maximum-treedepth hits, and E-BFMI.[30] The prespecified numerical checks were $\hat{R} < 1.01$, bulk and tail ESS of at least 400 for every sampled model parameter, no divergences, no maximum-treedepth hits, and chain-specific E-BFMI of at least 0.30.

Table S8: Posterior summaries from the Stan aggregate arm-level models. Entries are posterior medians with central 95% credible or predictive intervals.

Quantity	Gaussian RE	Standardized t_4 RE
Pooled PF_+	0.433 [0.222, 0.599]	0.421 [0.233, 0.580]
μ	-0.568 [-0.914, -0.251]	-0.547 [-0.868, -0.265]
τ	0.385 [0.171, 0.768]	0.434 [0.176, 0.872]
Predictive $PF_{+,new}$	0.430 [-0.436, 0.788]	0.422 [-0.584, 0.794]
$\Pr(PF_{+,new} > 0)$	0.908	0.909
$\Pr(PF_{+,new} > 0.20)$	0.802	0.813

The pooled center is similar under the two random-effects distributions. The Gaussian model gives a posterior median prevented fraction of 0.433, while the standardized t_4 model gives 0.421, a difference of only about 1.2 percentage points. The posterior median heterogeneity is somewhat larger under the robust model, 0.434 versus 0.385, and its upper credible limit is also larger. The principal difference therefore appears in prediction rather than in the pooled historical center. The lower endpoint of the 95% predictive interval changes from -0.436 under Gaussian random effects to -0.584 under the heavier-tailed t_4 model, while the upper endpoints remain similar.

Relative to the contrast-based normal-normal analysis in the main paper, the aggregate arm-level Gaussian model gives a somewhat larger pooled historical prevented fraction. This difference is consistent with the arm-level model serving its intended role: it tests whether the substantive conclusion depends on first constructing a log ratio and plug-in delta-method sampling variance. The direction of the historical effect is unchanged, but the arm-level model admits somewhat larger heterogeneity and correspondingly broader prediction.

S4.5 Hamiltonian Monte Carlo diagnostics

The computational diagnostics were uniformly satisfactory. The detailed results are shown in Table S9.

Table S9: Stan convergence and Hamiltonian Monte Carlo diagnostics. ESS values are minima over sampled model parameters; $\hat{R}$ is the corresponding maximum.

Diagnostic	Gaussian RE	Standardized t_4 RE
Maximum $\hat{R}$	1.000855	1.000939
Minimum bulk ESS	5356	4562
Minimum tail ESS	6391	5112
Largest MCSE(mean), reported targets	0.0034	0.0057
Divergent transitions	0	0
Maximum-treedepth hits	0	0
Minimum E-BFMI	0.767	0.683

Both models therefore satisfy the prespecified convergence criteria by wide margins. No divergent transitions or maximum-treedepth hits occurred, and the minimum E-BFMI was well above the threshold used to flag problematic exploration of the Hamiltonian energy distribution. The minimum bulk and tail ESS values were both above 4500 despite only 16,000 retained draws per model, and the maximum rank-normalized $\hat{R}$ was below 1.001. The CmdStan diagnostic routine reported no problems for either fit.

S4.6 Posterior predictive assessment

For every retained posterior draw, replicated treatment and control sample means were generated from the fitted arm-level sampling model. Two global discrepancy statistics were used. The first was the sum of squared standardized arm-level residuals,

$$T_{SS} = \sum_{i=1}^9 \left\{ \left(\frac{\bar{X}_{Ci} - m_i}{s_{Ci}/\sqrt{n_{Ci}}} \right)^2 + \left(\frac{\bar{X}_{Ti} - m_i e^{\theta_i}}{s_{Ti}/\sqrt{n_{Ti}}} \right)^2 \right\}.$$

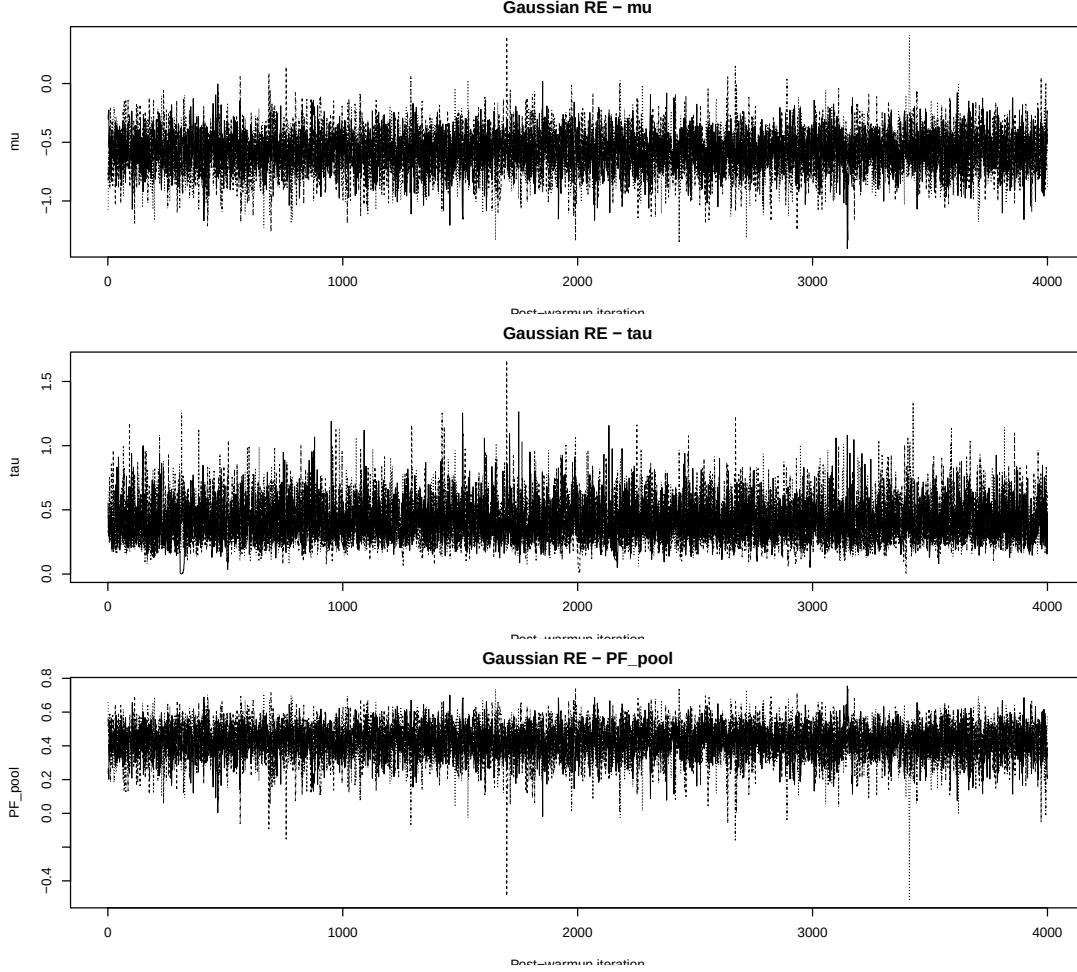


Figure S6: Post-warm-up trace plots for μ , τ , and the pooled prevented fraction under the Gaussian random-effects arm-level model.

The second was the maximum absolute standardized arm-level residual,

$$T_{\max} = \max_{i, a \in \{C, T\}} |r_{ia}|.$$

For each discrepancy, the posterior predictive tail area was calculated as the fraction of draws for which the replicated statistic exceeded the corresponding observed statistic.

The two global posterior predictive tail areas are close to one half under both models. Thus neither the overall arm-level dispersion nor the largest standardized residual appears unusual relative to replicated datasets from the fitted models. These checks do not establish that the aggregate Gaussian sampling approximation is correct, but they provide no indication of a gross mismatch on the two discrepancies examined.

Milsom is of particular interest because the frequentist leave-one-study-out analysis shows that it influences heterogeneity and the prediction interval. Under the Gaussian arm-level model, its study-specific posterior prevented fraction has median 0.065 and 95% credible interval $[-0.287, 0.330]$. Under the standardized t_4 model, the corresponding summary is 0.060 $[-0.308, 0.333]$. The posterior probability that a fresh standardized random effect is at least as extreme in absolute value as

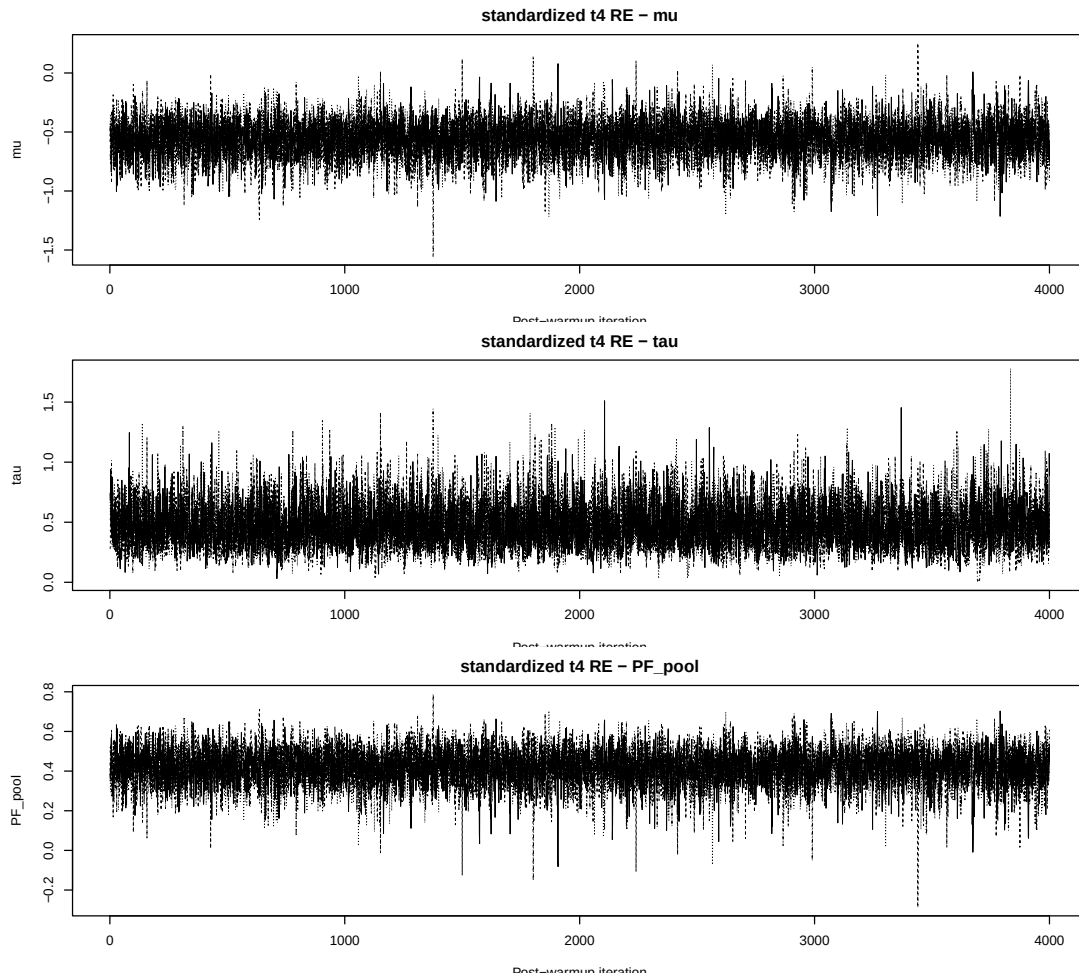


Figure S7: Post-warm-up trace plots for μ , τ , and the pooled prevented fraction under the standardized t_4 random-effects arm-level model.

Milsom's fitted standardized effect is approximately 0.25 in both models. Thus Milsom remains the study most compatible with a null or adverse true effect, but the robust model does not treat it as an extraordinarily remote draw from the fitted random-effects population.

S4.7 Study-specific posterior effects

Table S11 reports the study-specific posterior prevented fractions from the two arm-level models. The robust specification has little effect on the posterior centers of the well-estimated studies. Larger differences appear mainly in the tails for studies with less precise or more extreme effects.

S4.8 Robustness of the historical center and transportability

The production fits sharpen the distinction between the historical average and transportability. The two arm-level models differ little in their pooled posterior centers, but the standardized t_4 specification assigns more mass to large between-study departures. This is reflected in the posterior

Table S10: Posterior predictive checks for the aggregate arm-level models. Values close to zero or one would place the observed discrepancy in a tail of the replicated distribution.

Check	Gaussian RE	Standardized t_4 RE
$\Pr(T_{SS}^{\text{rep}} \geq T_{SS}^{\text{obs}} \mid \mathcal{D})$	0.434	0.427
$\Pr(T_{\text{max}}^{\text{rep}} \geq T_{\text{max}}^{\text{obs}} \mid \mathcal{D})$	0.456	0.445
$\Pr(z_{\text{new}} \geq z_{\text{Milsom}} \mid \mathcal{D})$	0.253	0.249

Table S11: Study-specific posterior prevented fractions under the two aggregate arm-level models. Entries are posterior medians with central 95% credible intervals.

Study	Gaussian RE	Standardized t_4 RE
Koch	0.632 [0.395, 0.828]	0.629 [0.384, 0.863]
Modeer	0.348 [0.118, 0.530]	0.355 [0.127, 0.530]
Clark	0.243 [0.060, 0.393]	0.249 [0.066, 0.396]
Tewari	0.625 [0.401, 0.817]	0.623 [0.390, 0.841]
Bravo	0.429 [0.285, 0.554]	0.428 [0.287, 0.552]
Sköld	0.545 [0.361, 0.688]	0.534 [0.353, 0.682]
Arruda	0.406 [0.260, 0.526]	0.406 [0.262, 0.525]
Tagliaferro	0.433 [−0.024, 0.711]	0.428 [−0.012, 0.700]
Milsom	0.065 [−0.287, 0.330]	0.060 [−0.308, 0.333]

median of τ , its upper credible tail, and especially the lower tail of the posterior predictive prevented fraction. The robust analysis therefore changes the assessment of extreme future-study behavior more than it changes the estimated historical mean effect.

The posterior probabilities of any preventive effect in a new study are almost identical, 0.908 under Gaussian random effects and 0.909 under standardized t_4 random effects. The corresponding probabilities of at least a 20% preventive fraction are 0.802 and 0.813. These probabilities summarize the entire predictive distribution but should not obscure its width: both 95% predictive intervals extend across zero and well into the adverse side.

S4.9 Reproducibility

The production analysis used R 4.5.3, CmdStanR 0.9.0, posterior 1.7.0, and CmdStan 2.40.0. The Gaussian and standardized t_4 models used random-number seeds 20260927 and 20260928, respectively. The archived production directory contains the original CmdStan CSV files for all eight chains, complete parameter summaries, HMC diagnostic output, posterior predictive summaries, figures, the frozen input data, and the R session information.

The complete production diagnostics from CmdStan reported satisfactory treedepth, no divergent transitions, satisfactory E-BFMI, satisfactory rank-normalized split effective sample sizes, and satisfactory rank-normalized split $\hat{R}$ values for both models. Consequently, the numerical differences between the Gaussian and standardized t_4 fits should be interpreted as model differences rather than as visible Monte Carlo convergence problems.

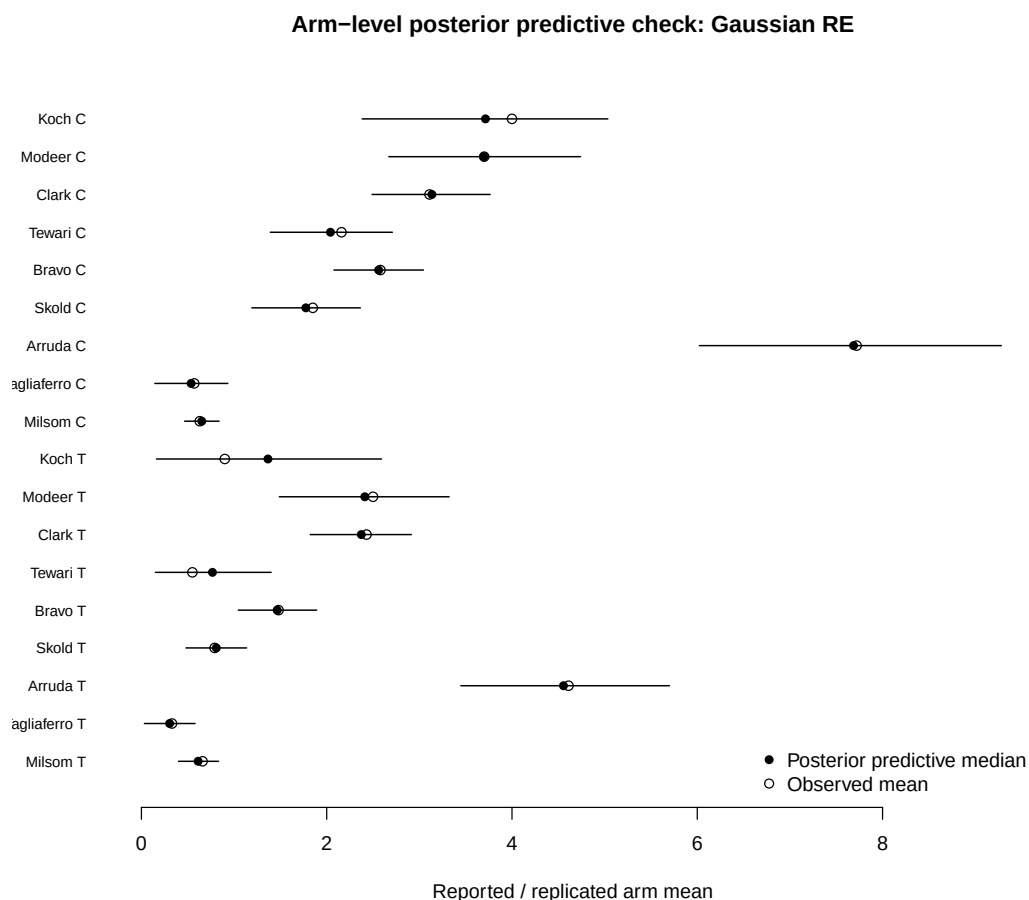


Figure S8: Arm-level posterior predictive intervals under Gaussian random effects. Open points are the observed arm means; filled points and horizontal intervals are posterior predictive medians and central 95% intervals.

S5 Contemporary Continuous Outcomes and Temporal-Model Sensitivity

The primary calendar-time analysis uses 14 continuous-outcome comparisons: the nine historical studies plus Muñoz-Millán, Latifi-Xhemajli, McMahon, Zeng, and Wang.[23, 19, 22, 33, 31] Jayasinghe and Ghasemi are not assigned primary inverse-variance weights because their published summaries do not determine a unique design-adjusted sampling variance.[17, 10]

S5.1 Contemporary extraction and variance audit

Table S12 records the contemporary continuous-outcome inputs and the treatment of their sampling variances. The purpose of this audit is to distinguish quantities that are directly supported by arm-level summaries from quantities that require clustering or overdispersion assumptions.

For Wang, replacing the planning cluster size $m = 40$ by the observed mean completed cluster size

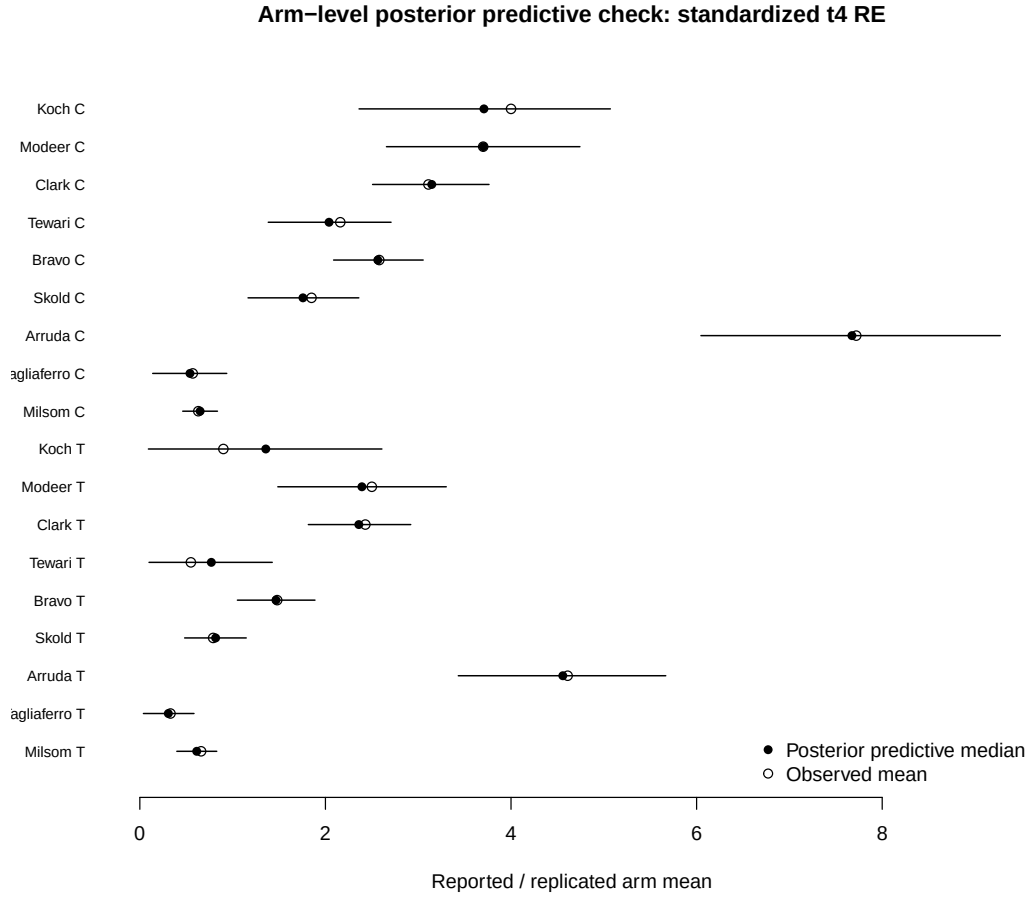


Figure S9: Arm-level posterior predictive intervals under standardized t_4 random effects.

5005/107 changes the fitted calendar slope only from 0.0986 to approximately 0.0992 per decade. This sensitivity supports the use of the planning design effect for the primary analysis, while the qualifier “approximately cluster-adjusted” remains important.

S5.2 Alternative temporal functional forms

Let

$$t_i = \frac{\text{Year}_i - 2000}{10}.$$

The linear random-effects model is

$$y_i = \beta_0 + \beta_1 t_i + u_i + \epsilon_i, \quad u_i \sim N(0, \tau^2), \quad \epsilon_i \sim N(0, s_i^2).$$

Positive β_1 means that the treatment-to-control ratio moves toward one in later studies and hence that the incremental prevented fraction attenuates.

Because the candidate models use different fixed-effect design matrices, comparison is based on maximum-likelihood AICc rather than restricted likelihoods.

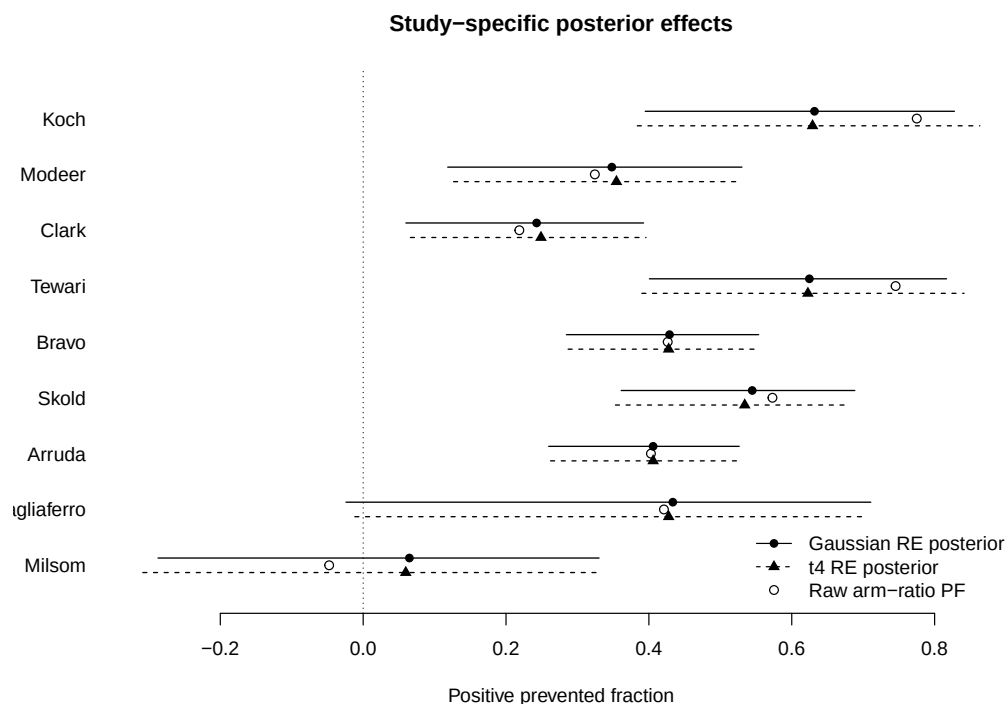


Figure S10: Study-specific posterior prevented fractions from the Gaussian and standardized t_4 arm-level models, together with the raw prevented fractions calculated from the reported arm means.

The intercept-only model has the smallest AICc, so the data do not require a calendar-time moderator. The linear-time and historical-versus-contemporary models are nevertheless both within two AICc units of the no-trend model. Among specifications that permit evolution over time, the linear model is the most parsimonious; the quadratic and three-era models do not gain sufficient fit to offset their additional parameters.

S5.3 Primary linear-time fit and era comparison

The REML–Hartung–Knapp linear fit gives

$$\hat{\beta}_0 = -0.4768, \quad \hat{\beta}_1 = 0.0986 \text{ per decade}, \quad \hat{\tau} = 0.2787.$$

The 95% Hartung–Knapp interval for the time coefficient is

$$[-0.0445, 0.2417].$$

On the ratio scale,

$$\exp(0.0986) = 1.104,$$

so the fitted treatment-to-control ratio is multiplied by about 1.10 for each later decade. The corresponding fitted prevented fraction decreases from approximately 34.8% in 2005 to 20.6% in 2025. These are fitted cross-study values, not repeated measurements of one population.

Without imposing a linear trajectory, pooling the five contemporary primary studies gives a REML center of approximately 22.7% on the positive prevented-fraction scale, with a Hartung–Knapp

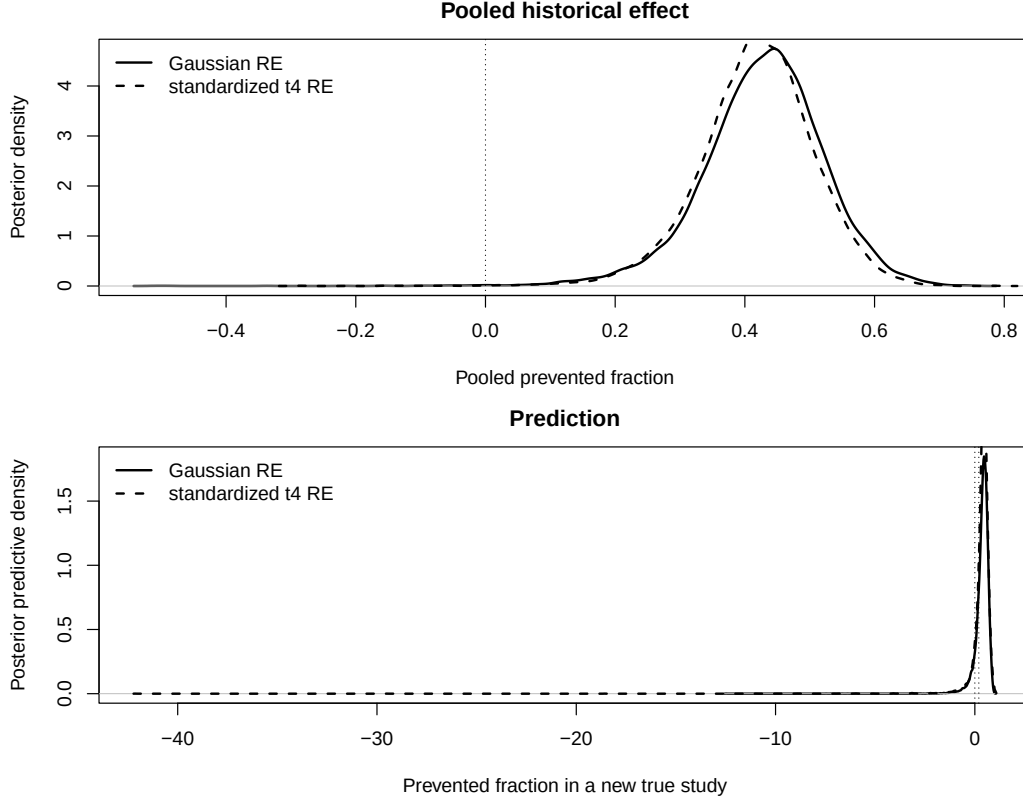


Figure S11: Posterior distributions of the pooled prevented fraction (top) and posterior predictive distributions for a new true study effect (bottom) under Gaussian and standardized t_4 random effects.

interval of approximately $[-12.9\%, 47.1\%]$. The corresponding contemporary-minus-historical era coefficient is

$$\hat{\beta}_E = 0.242,$$

with 95% Hartung–Knapp interval

$$[-0.190, 0.675].$$

Thus the historical and contemporary fitted centers differ materially at the point-estimate level, but the contrast remains imprecisely estimated.

The Bayesian calendar-time model uses

$$\beta_0 \sim N(0, 1^2), \quad \beta_1 \sim N(0, 0.25^2), \quad \tau \sim \text{Half-Normal}(0, 0.5^2),$$

and gives

$$\text{median}(\beta_1 \mid \mathcal{D}) = 0.091, \quad 95\% \text{ CrI} = [-0.036, 0.231],$$

with

$$\Pr(\beta_1 > 0 \mid \mathcal{D}) = 0.924.$$

The frequentist and Bayesian analyses therefore agree on the direction of the point estimate while retaining substantial uncertainty about its magnitude.

Table S12: Contemporary continuous-outcome variance audit. Means are followed by SDs in parentheses.

Study	Status	Endpoint	Treatment vs control	Variance treatment / audit note
Muñoz-Millán	Primary	24-month dmft	1.6(2.4), $n = 131$ vs 2.1(2.5), $n = 144$	Direct first-order LRR variance. Primary-article SDs 2.4/2.5 are used in place of a conflicting secondary extraction.
Latifi-Xhemajli	Primary	24-month dmfs	5.2(10.5), $n = 218$ vs 10.1(12.9), $n = 209$	Direct first-order LRR variance. The quantity is endpoint burden rather than a pure increment and is therefore also an endpoint-sensitivity case.
McMahon	Primary	24-month d_3 mfs	3.5(5.9), $n = 577$ vs 3.5(4.9), $n = 573$	Direct first-order LRR variance. The primary trial gives identical final means; SDs are taken from the evidence extraction used in the production dataset.
Zeng	Primary	24-month full-mouth Δ DMFT/dmft	2.05(2.64), $n = 79$ vs 1.96(2.56), $n = 80$	Direct first-order LRR variance for TFV+OHE versus OHE.
Wang	Primary; approximate cluster adjustment	24-month first-permanent-molar DFS increment	0.38(1.21), $n = 2385$ vs 0.61(1.60), $n = 2620$	Naive LRR variance multiplied by the design effect $1 + (40 - 1)(0.20) = 8.8$, using the published ICC and planning cluster size. No directly reported cluster-robust LRR SE is available.
Jayasinghe	Sensitivity only	2-year mean number of new caries	1.50, $n = 162$ vs 1.97, $n = 161$	No usable SD or SE is reported for the two-year mean; uncertainty is explored over a quasi-Poisson overdispersion grid.
Ghasemi	Sensitivity only	6-month DMFS increment	0.2(0.77), $n = 173$ vs 0.8(1.59), $n = 179$	Only four schools were randomized and no outcome ICC or school-specific increment summaries are reported; uncertainty is explored over an ICC/design-effect grid.

S5.4 Influence and variance sensitivity

Deleting each contemporary primary study in turn gives the following REML–Hartung–Knapp slopes.

Every deletion leaves a positive point estimate. Adding Jayasinghe under a representative quasi-Poisson overdispersion assumption gives a slope near 0.094 per decade. Adding Ghasemi as well under a representative ICC of 0.05 gives a slope near 0.085. Across the joint Jayasinghe/Ghasemi variance grid, the fitted slope remains positive, approximately 0.071–0.091 per decade. These calculations are deliberately treated as variance sensitivities rather than substitutes for unavailable design-adjusted sampling information.

S6 Contemporary Binary Outcomes as Supporting Evidence

Several contemporary trials report child-level caries incidence rather than a continuous DMF-type outcome. These studies estimate a different quantity from a ratio of mean caries outcomes and are

Table S13: Maximum-likelihood comparison of temporal specifications for the 14-study primary continuous-outcome dataset. Akaike weights are descriptive rather than posterior model probabilities.

Model	AICc	Δ AICc	Akaike weight
No calendar moderator	17.29	0.00	0.430
Linear calendar time	18.13	0.84	0.283
Historical vs contemporary era	18.73	1.44	0.209
Quadratic calendar time	21.73	4.44	0.047
Three calendar eras	22.58	5.29	0.031

Table S14: Leave-one-contemporary-study-out sensitivity for the 14-study calendar-time model.

Omitted study	$\hat{\beta}_1$ per decade	95% HK interval	$\hat{\tau}$
Muñoz-Millán	0.101	$[-0.055, 0.257]$	0.301
Latifi-Xhemajli	0.116	$[-0.029, 0.261]$	0.264
McMahon	0.076	$[-0.074, 0.227]$	0.273
Zeng	0.079	$[-0.077, 0.235]$	0.282
Wang	0.112	$[-0.043, 0.266]$	0.291

therefore synthesized separately. For study i we use

$$r_i = \log \left(\frac{p_{Ti}}{p_{Ci}} \right),$$

where $r_i < 0$ favors a varnish-containing strategy.

Table S15: Contemporary binary comparisons used as supporting evidence. The reported values are crude risk ratios and are not combined with the continuous-outcome synthesis.

Study	Binary endpoint / contrast	Year	Crude RR
Muñoz-Millán	New caries; fluoride varnish versus placebo	2018	0.81
McMahon	Worsening d_3 mft; varnish plus Childsmile treatment-as-usual versus treatment-as-usual	2020	0.85
Wang	First-permanent-molar caries incidence; varnish versus no varnish	2022	0.72
Zeng	PFM caries incidence; TFV+OHE versus OHE	2025	1.08
He	New cavitated primary-molar caries; 5% NaF varnish versus placebo	2026	0.81

The five study descriptions correspond to Muñoz-Millán, McMahon, Wang, Zeng, and He. [23, 22, 31, 33, 13] An exploratory REML random-effects synthesis on the log-risk-ratio scale gives

$$\hat{\rho} = -0.225, \quad \hat{\tau}_R = 0.073,$$

or

$$\exp(\hat{\rho}) \approx 0.80.$$

The approximate 95% interval for the pooled risk ratio is

$$[0.71, 0.89].$$

Under a Half-Normal(0, 0.5²) heterogeneity prior, the posterior median risk ratio is also approximately 0.80, with 95% credible interval near [0.69, 1.00]. The corresponding posterior predictive interval is much wider, approximately

$$[0.56, 1.24].$$

This synthesis is intentionally described as supporting evidence. Several trials have clustered, multilevel, or otherwise complex designs, and crude event ratios do not replace the design-adjusted analyses reported in the individual publications. In addition, several studies contribute both a continuous and a binary endpoint, so the two evidence streams are not independent replications. Their numerical agreement is informative as a cross-outcome check, but the estimands are not pooled into a common effect.

S7 Explanatory Meta-Regression for the Calendar-Time Signal

Calendar year is a study-level marker of secular change rather than a clinical mechanism. We therefore ask whether the estimated calendar coefficient is reduced after introducing two measurable study characteristics connected to the changing counterfactual condition: organized background prevention and control-arm caries burden. The final analysis uses the same 14-study primary continuous-outcome dataset as the calendar-time model, including the approximately cluster-adjusted Wang comparison.

S7.1 Coding of organized background prevention

We define

$$B_i = \begin{cases} 1, & \text{the control arm received a protocol-specified, study-organized} \\ & \text{non-varnish preventive intervention in addition to ordinary care,} \\ 0, & \text{otherwise.} \end{cases}$$

Routine community exposure to fluoridated toothpaste or water, when not delivered as a protocol-specified control-arm intervention, is not by itself sufficient for $B_i = 1$. Under this rule, $B_i = 1$ for Tagliaferro, Muñoz-Millán, McMahon, and Zeng. Their control conditions included, respectively, repeated oral-health education, provision of toothpaste and toothbrushes during follow-up, the multicomponent Childsmile treatment-as-usual program, and oral-health education. [?, 23, 22, 33] Wang and the remaining studies are coded $B_i = 0$ under this narrow definition.

The binary coding is intentionally low-dimensional. It distinguishes an explicitly organized preventive background from ordinary or unspecified care; it does not imply that all $B_i = 1$ interventions have equal intensity.

S7.2 Bayesian model sequence

Let

$$t_i = \frac{\text{Year}_i - 2000}{10}.$$

Table S16: Moderator coding used in the final 14-study explanatory meta-regression. The control mean is the quantity entering the continuous log-ratio analysis and is used only as an exploratory burden proxy.

Study	Year	B_i	Control-arm mean
Koch	1975	0	4.00
Modeer	1984	0	3.70
Clark	1985	0	3.11
Tewari	1990	0	2.16
Bravo	1997	0	2.58
Sköld	2005	0	1.85
Tagliaferro	2011	1	0.57
Milsom	2011	0	0.63
Arruda	2012	0	7.72
Muñoz-Millán	2018	1	2.10
Latifi-Xhemajli	2019	0	10.10
McMahon	2020	1	3.50
Wang	2022	0	0.61
Zeng	2025	1	1.96

The time-only model is

$$\theta_i \sim N(\beta_0 + \beta_1 t_i, \tau^2),$$

and the organized-prevention model is

$$\theta_i \sim N(\beta_0 + \beta_1 t_i + \beta_2 B_i, \tau^2).$$

Positive β_1 denotes movement of the treatment-to-control ratio toward one over calendar time. Positive β_2 means that, conditional on calendar time, a protocol-specified preventive control condition is associated with a smaller incremental varnish contrast.

The priors are

$$\beta_0 \sim N(0, 1^2), \quad \beta_j \sim N(0, 0.25^2), \quad j = 1, 2, 3,$$

and

$$\tau \sim \text{Half-Normal}(0, 0.5^2).$$

S7.3 Control-arm burden and the shared-denominator problem

A direct regression of $y_i = \log(\bar{X}_{Ti}/\bar{X}_{Ci})$ on the observed control mean is problematic because the control mean is estimated with sampling error and also appears in the response denominator. Ordinary meta-regression can therefore induce a mathematical association even in the absence of genuine effect modification.[28, 1]

For the burden-adjusted sensitivity analysis, define

$$x_{Ti} = \log(\bar{X}_{Ti}), \quad x_{Ci} = \log(\bar{X}_{Ci}),$$

with first-order sampling variances

$$v_{Ti} = \frac{s_{Ti}^2}{n_{Ti}\bar{X}_{Ti}^2}, \quad v_{Ci} = \frac{s_{Ci}^2}{n_{Ci}\bar{X}_{Ci}^2}.$$

Let a_i be the latent true log control-arm mean. The joint arm approximation is

$$\begin{aligned} x_{Ci} \mid a_i &\sim N(a_i, v_{Ci}), \\ x_{Ti} \mid a_i, \theta_i &\sim N(a_i + \theta_i, v_{Ti}), \end{aligned}$$

with

$$\theta_i \sim N \left[\beta_0 + \beta_1 t_i + \beta_2 B_i + \beta_3 \left(\frac{a_i - \bar{x}_C}{s_{\log C}} \right), \tau^2 \right].$$

Thus β_3 is the change in true log ratio associated with a one-standard-deviation increase in latent log control-arm burden. This formulation propagates uncertainty in the control mean and respects its shared role in the response and moderator.

S7.4 Posterior results

Table S17: Bayesian explanatory meta-regression for the 14-study primary continuous-outcome dataset. Intervals are central 95% credible intervals.

Model	Calendar coefficient β_1	$\Pr(\beta_1 > 0 \mid \mathcal{D})$	Additional moderator
Time only	0.091 [−0.036, 0.231]	0.924	–
Time + organized prevention	0.057 [−0.071, 0.213]	0.804	$\beta_2 = 0.178$ [−0.195, 0.505]
Time + organized prevention + burden	0.053 [−0.074, 0.212]	0.789	$\beta_2 = 0.171$ [−0.205, 0.498]; $\beta_3 = -0.060$ [−0.240, 0.135]

Adding organized background prevention reduces the posterior median calendar coefficient from 0.091 to 0.057 per decade, a reduction of approximately 37%. The background-prevention coefficient has posterior median 0.178 and

$$\Pr(\beta_2 > 0 \mid \mathcal{D}) = 0.834.$$

Thus its posterior direction is consistent with a smaller incremental varnish effect when an organized preventive program is already present, but its credible interval remains wide.

Adding the error-aware burden term changes the calendar coefficient only slightly further, from 0.057 to 0.053. The burden coefficient has median -0.060 with 95% CrI [−0.240, 0.135] and $\Pr(\beta_3 < 0 \mid \mathcal{D}) \approx 0.75$. Its direction is compatible with a larger relative varnish benefit in higher-burden control populations, but the available data do not estimate that relation precisely.

The explanatory models therefore sharpen rather than overturn the temporal interpretation. Measurable strengthening of the preventive environment accounts for a non-negligible portion of the fitted time signal, while substantial uncertainty remains. These are observational study-level moderator analyses, not causal mediation analyses: preventive intensity and calendar time are correlated, the number of studies is small, and the burden proxy mixes teeth, surfaces, endpoint burden, and increments.

S8 Reproducibility and Final Analysis Record

The Supplement is intended to provide a technical audit trail rather than a second set of competing primary analyses. The production workflow uses a frozen set of study-level inputs and keeps the

historical, temporal, binary, and moderator analyses in separate data objects so that changes in one evidence stream do not silently alter another.

The historical REML–Hartung–Knapp benchmark was generated in R 4.5.3 using `metafor` 5.2-1. The Bayesian normal–normal prior-sensitivity analysis uses deterministic numerical integration through `bayesmeta`, avoiding Monte Carlo error in that model. The aggregate arm-level Gaussian and standardized t_4 models were fit with CmdStanR 0.9.0, `posterior` 1.7.0, and CmdStan 2.40.0. The two Stan fits used random-number seeds 20260927 and 20260928, respectively. The temporal production workflow uses `metafor` for the frequentist models and a separate Stan implementation for the Bayesian calendar-time analysis.

The frozen production archive should contain the arm-level historical data, the Cochrane-13 reconstruction, the contemporary variance-audit table, the primary 14-study temporal dataset, all Jayasinghe/Ghasemi sensitivity scenarios, the binary-outcome extraction, the moderator coding, R and Stan source files, machine-readable output tables, session information, random seeds, and the final figure files. Numerical values in the manuscript should be generated from those archived outputs rather than re-entered manually.

The role of these supplementary analyses is consistent across sections: the historical pooled center is tested against alternative effect constructions, heterogeneity priors, random-effects distributions, individual-study deletions, and completion of the Cochrane historical set; the temporal conclusion is tested against alternative functional forms, contemporary-study deletions, and uncertain modern variances. The resulting evidence is strongest for the stability of the historical average, while prediction and the magnitude of temporal attenuation remain appropriately more uncertain.

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